TP-Link Omada EAP772-Outdoor technical requirements

For your case — **TP-Link Omada EAP772-Outdoor as AP + Wi-Fi dongle clients in smart factory processing line** — this is the practical technical package.

1. Main technical specifications

Item	TP-Link Omada EAP772-Outdoor
Wi-Fi standard	Wi-Fi 7 / IEEE 802.11be/ax/ac/n/g/b/a
Bands	2.4 GHz, 5 GHz, 6 GHz
Max wireless rate	Up to **10,777 Mbps** total: 688 Mbps + 4,324 Mbps + 5,765 Mbps
Ethernet	**1× 2.5GbE RJ45 PoE+ port**
Power	802.3at PoE
Max power consumption	**21.5 W**
Antennas	2.4 GHz: 2×4 dBi, 5 GHz: 2×6 dBi, 6 GHz: 2×6 dBi
Weatherproof	**IP68**
Roaming / mesh	Omada Mesh and Seamless Roaming support
Central management	Omada Cloud-Based Controller, Hardware Controller, Software Controller, Omada App
Official connected-device capacity	**380+ connected devices**
Official recommended coverage	**300 m² / 3,200 ft²**

TP-Link's official page lists the EAP772-Outdoor as a tri-band outdoor Wi-Fi 7 AP with 300 m² recommended omnidirectional coverage, 380+ connected devices, IP68 protection, 1×2.5GbE PoE+ port, and Omada mesh/roaming/centralized management support. (Omada)

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2. Radius of this AP

Officially, TP-Link gives **coverage area**, not radius: **300 m² / 3,200 ft² recommended coverage**. (Omada)

If we convert 300 m² into a circular coverage radius:

```
Radius = √(300 / π) ≈ 9.8 m
```

So, the **official recommended coverage radius equivalent is around 10 m**.

For a real smart factory, I would use this planning rule:

Environment	Practical planning radius
Open outdoor / clean open space	~10–20 m reliable design radius
Indoor smart factory with metal machines	~8–15 m reliable design radius
AGV / moving dongle client route	Design AP spacing around **15–25 m** with overlap
High-reliability processing line	Keep RSSI better than **-67 dBm** across the full route

Do not design based on maximum possible Wi-Fi distance. For factory mobility, design based on **stable RSSI, low packet loss, and AP overlap**.

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3. How many dongle clients can work on one AP?

Officially, EAP772-Outdoor supports **380+ connected devices**. (Omada)

For your smart-factory dongle-client case, use a much more conservative engineering number:

Client type	Recommended dongle clients per AP
Simple sensor/status dongles	50–100 possible
AGV / processing cart telemetry	**20–50 recommended**
Moving AGV/cart with roaming	**10–30 recommended**

Client type	Recommended dongle clients per AP
Dongle clients with video/camera upload	**5–15 recommended**
Critical process control	Keep very conservative; do not use normal Wi-Fi dongle for safety-critical control

For your simulation, I recommend planning **10–20 mobile dongle clients per AP** first. After testing channel utilization, packet loss, and roaming, you can increase the number.

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4. Official manual / tutorial / documentation package

Document / resource	Use it for
EAP772-Outdoor Product Page	Main specs: Wi-Fi 7, coverage, connected devices, PoE, antennas, IP68, roaming, mesh
EAP772-Outdoor Support Page	Central place for datasheet, hardware installation guide, reference guide, firmware, software, MIB files, and configuration guides
Hardware Installation Guide	Wall/pole mounting, physical installation, PoE cable installation
Datasheet	Technical specs, antenna gain, coverage, client count, power, wireless features
Reference Guide	Advanced configuration reference
Standalone Setup Guide	How to configure the AP without Omada Controller, using Web GUI or Omada App
Upgrade/Reboot/WLAN Schedule Guide	Firmware upgrade schedule, reboot schedule, Wi-Fi radio on/off schedule
Ethernet Cable / PoE Guide	Choosing proper cable for stable PoE power
Band Steering Guide	How to guide clients between 2.4/5/6 GHz bands
Firmware Download	Latest AP firmware
Software Download	Omada Controller, Discovery Utility, related tools

The official EAP772-Outdoor support page shows document categories including **Datasheet, Hardware Installation Guide, Reference Guide, Knowledgebase,**

Release Notes, and download categories including **Firmware, Software, MIBs, and GPL Code**. (Omada)

5. Software setting program

For your smart-factory AP deployment, use **Omada Controller**.

Software / tool	Purpose
Omada Software Controller	Free on-premises controller for centralized AP, switch, and gateway management
Omada Hardware Controller OC200 / OC300	Dedicated controller appliance; better for production than running software on a random PC
Omada Cloud-Based Controller	Cloud-managed controller option
Omada App	Mobile setup and monitoring
Omada Discovery Utility	Finds and adopts Omada APs across different subnets
Standalone Web GUI	Direct AP setup without controller; acceptable for one AP, not ideal for multiple APs
Firmware Download Center	Firmware update package for the AP

TP-Link describes Omada Controller as a free on-premises controller for Omada SDN that centrally manages access points, switches, and routers; it also lists the Omada Discovery Utility as a helper tool for managing EAPs in different subnets. (support.omadanetworks.com)

6. Recommended software setup for your simulation

For a **processing-line smart factory simulation**, use this setup:

```
Controller:
Omada Software Controller or OC200/OC300

AP:
3× TP-Link Omada EAP772-Outdoor

Client:
Industrial PC + Wi-Fi 6/7 dongle

SSID:
```

```
FACTORY_PROCESS_AGV

Band:
5 GHz first

Channel width:
20 MHz first, 40 MHz only if RF is clean

Security:
WPA2/WPA3-Personal for pilot
WPA2/WPA3-Enterprise for production

Roaming:
Enable Omada seamless roaming only after confirming dongle compatibility

IP/VLAN:
VLAN 20 for dongle clients
VLAN 10 for AP/controller management
```

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7. Basic tutorial: first setup process

Step 1 — Physical installation

```
PoE+ Switch → Shielded Cat6A cable → EAP772-Outdoor
```

Use a PoE+ switch because the AP supports **802.3at PoE** and has a **1×2.5GbE PoE+ RJ45 port**. (Omada)

Step 2 — Install controller

Use one of these:

```
Option A: Omada Software Controller on Windows/Linux server
Option B: OC200 or OC300 hardware controller
Option C: Omada Cloud-Based Controller
```

For factory use, I prefer **OC200/OC300** or a small dedicated mini-PC running **Omada Software Controller**.

Step 3 — Adopt APs

```
Omada Controller
→ Devices
→ Pending AP
→ Adopt
→ Assign AP name: AP-01, AP-02, AP-03
```

Step 4 — Create Wi-Fi network

```
Settings
→ Wireless Networks
→ Create New WLAN

SSID: FACTORY_PROCESS_AGV
Band: 5 GHz
Security: WPA2/WPA3
VLAN: 20
```

Step 5 — Configure radio

```
AP-01: Channel 36, 20 MHz, Medium power
AP-02: Channel 44, 20 MHz, Medium power
AP-03: Channel 149, 20 MHz, Medium power
```

Step 6 — Test dongle client

On the AGV/process PC:

```
Connect to FACTORY_PROCESS_AGV
Disable Wi-Fi power saving
Disable USB selective suspend
Check roaming behavior while moving
Run ping test to MES/process server
```

Step 7 — Validate factory route

```
Target RSSI: better than -67 dBm
Packet loss: near 0% for normal movement
```

```
Roaming interruption: should not break the application session
AP visibility: client should see at least 2 APs in overlap zones
```

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8. Recommended docs to download first

Download these first from the official support page:

```
1. EAP772-Outdoor Datasheet
2. EAP772-Outdoor Hardware Installation Guide
3. Omada Controller Software
4. Omada Discovery Utility
5. Latest EAP772-Outdoor Firmware
6. Standalone Mode Setup Guide
7. Upgrade/Reboot/WLAN Schedule Guide
8. Ethernet Cable / Stable PoE Guide
```

For your simulation, the most important are **Datasheet**, **Hardware Installation Guide**, **Omada Controller**, and **Firmware**. The support page also shows current EAP772-Outdoor firmware/release-note entries, including the 2026 builds for the US V1.6 hardware version. (Omada)